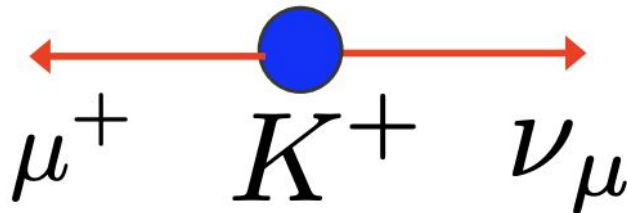
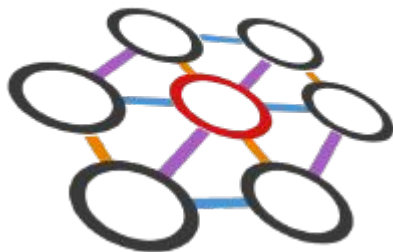


Wire-Cell KDAR

open data

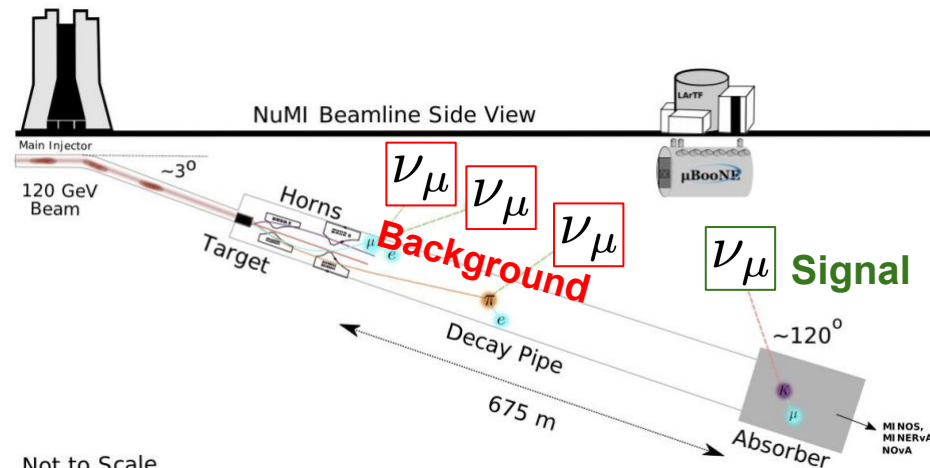
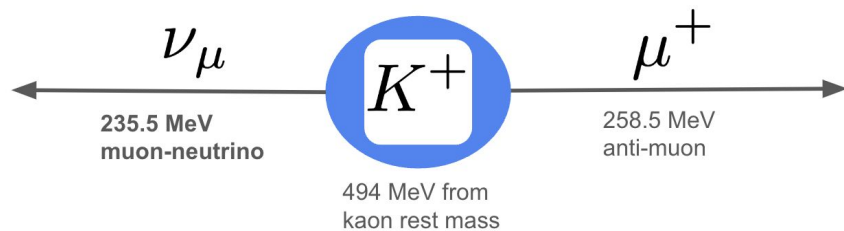
Benjamin Bogart
UbooNE XS Meeting
February 3rd, 2026



KDAR in MicroBooNE

Many more details in the [internal note](#)

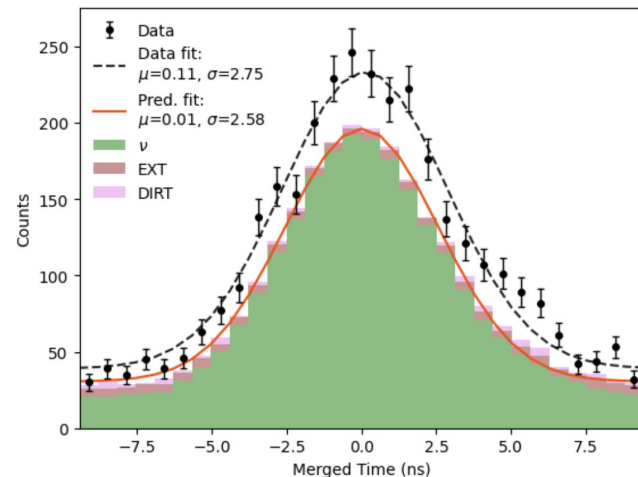
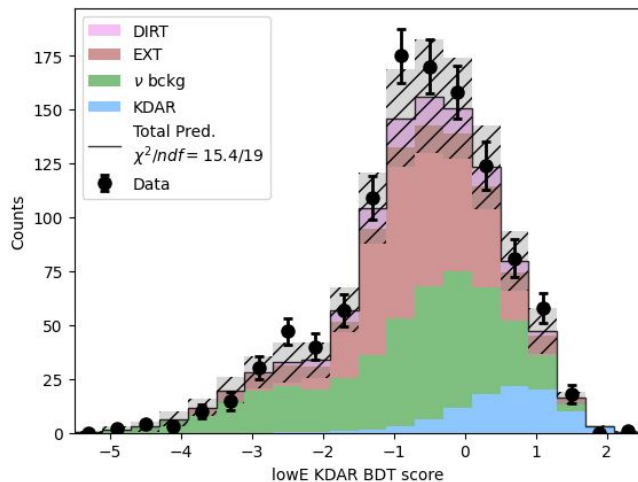
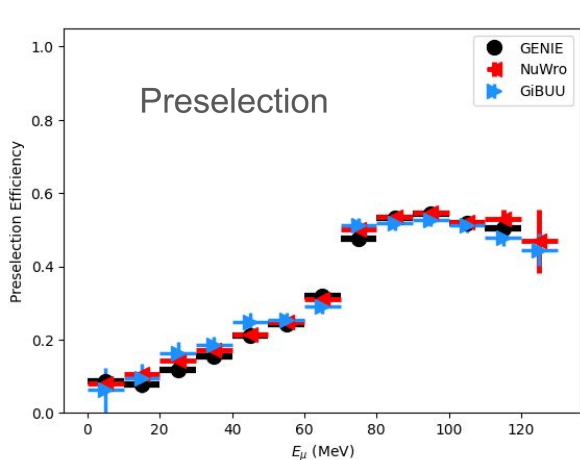
- Targeting a measurement of the 235.5 MeV monoenergetic KDAR flux originating from the NuMI absorber.
- Rare opportunity to study a monoenergetic neutrino source interacting with argon.
 - Utility for BSM analyses as well!



Analysis Strategy

Four stage selection:

- Modified Wire-Cell generic neutrino selection with improved acceptance of low energy events.
- Further preselection utilizing basic “no-brainer” cuts
 - Fully contained, reconstructed neutrino energy < 350 MeV, etc.
- BDT incorporating custom “tagger” variables to identify KDARs.
- ns timing cut to reject non-KDAR neutrinos in time with the beam bunches.



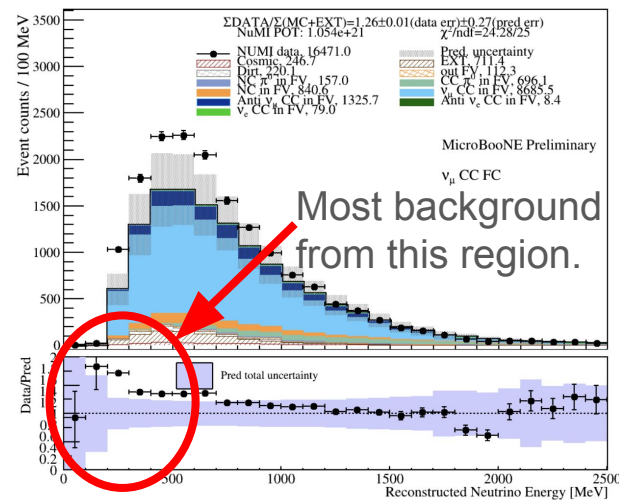
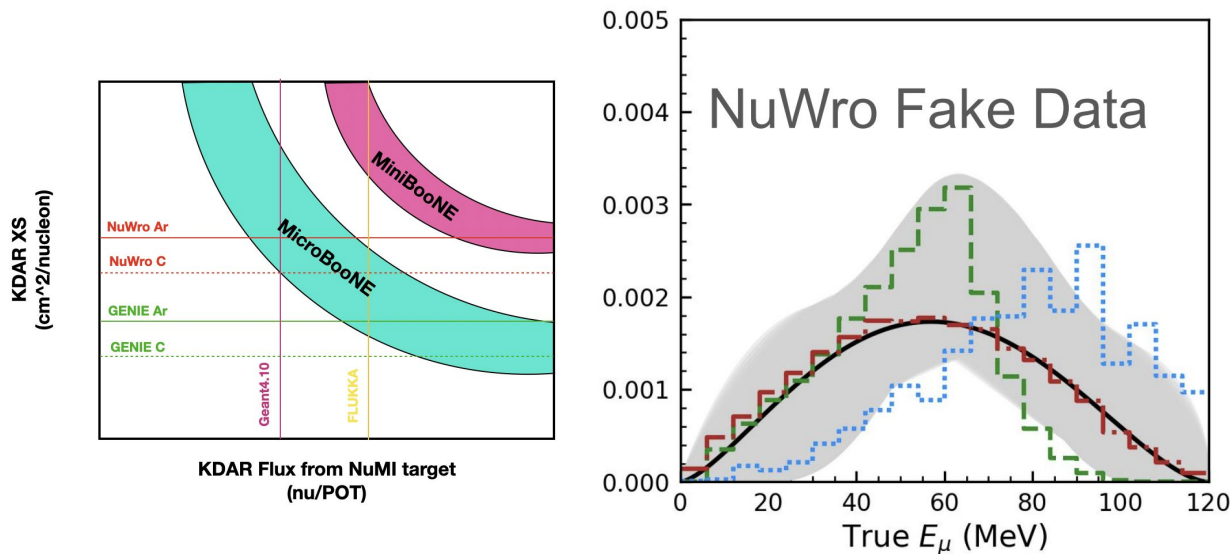
Analysis Strategy

Background constraint from numuCC selection:

- Helps account for the fact that the numuCC selection shows a data excess for NuMI.

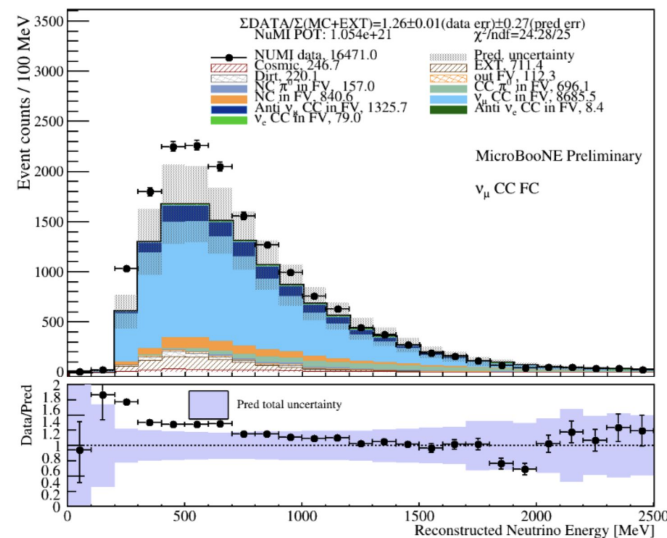
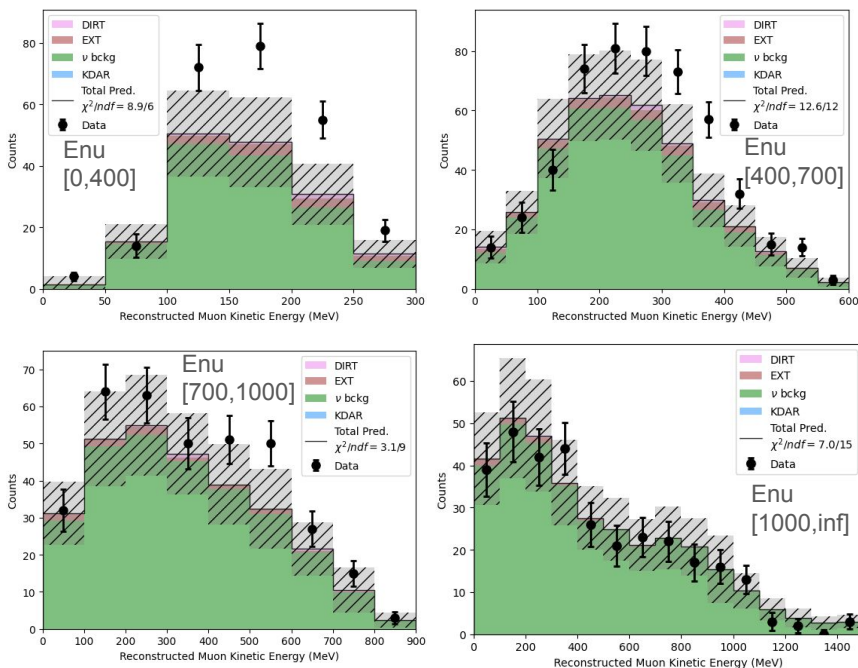
Report shape-only differential cross section and a “total rate” (flux times xs).

- Parameterized with a beta distribution rather than unfolded, R reported for chi2 comparisons.



Open data: sidebands

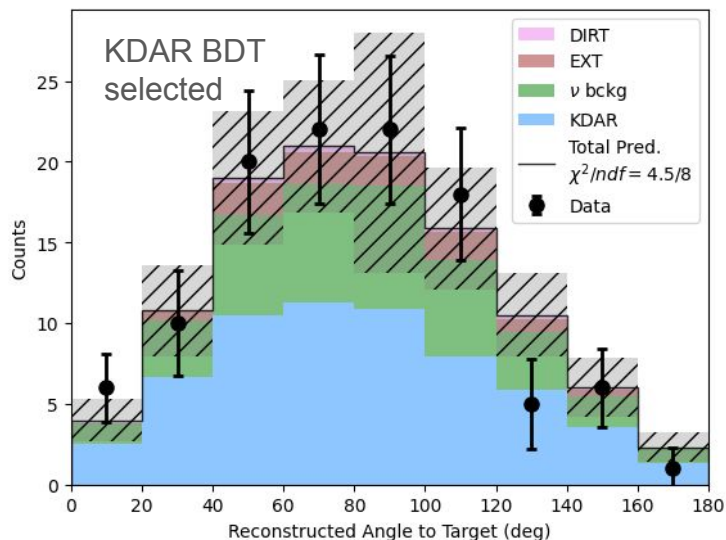
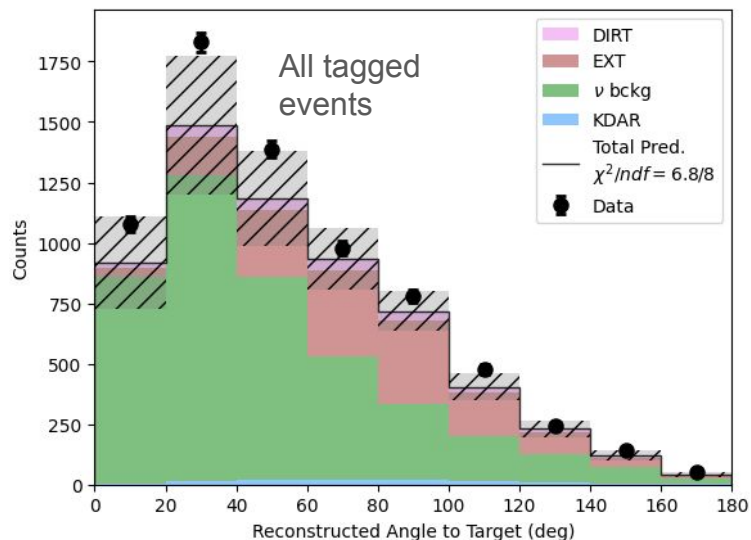
- Analysis has matured to the point of looking at open data.
- Utilizing 5e19 POT RHC and 5e19 POT FHC [open data samples](#).
- Know excess at low energies seen in the sideband numuCC distributions.
 - This creates a large pull in the background for the KDAR selection, effectively reducing the purity.



χ^2/ndf all bins = 35.5/42

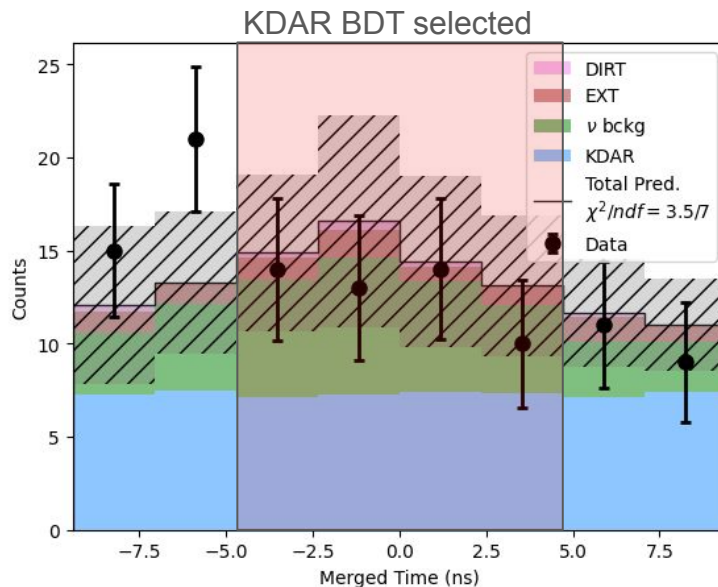
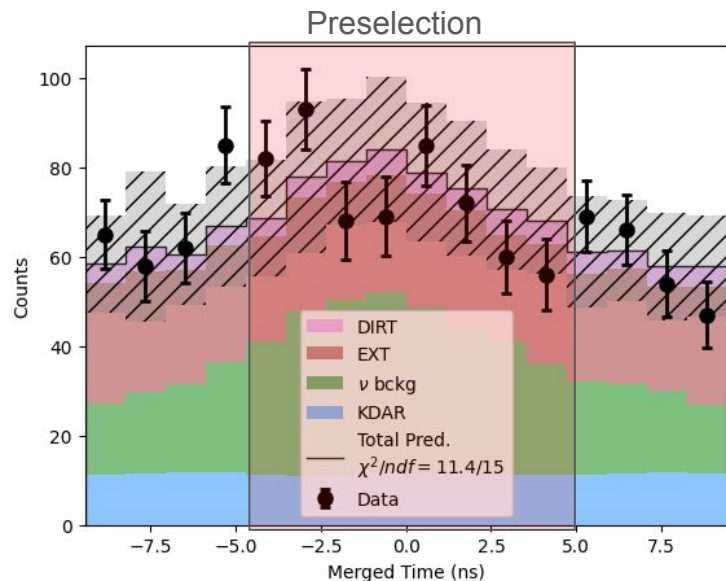
Open data: neutrino angular distributions

- The observed neutrino angular distribution is a good validation plot.
 - KDAR should be shifted away from the target relative to a more generic selection.
- A clear shift away from the target is seen when comparing all events with a tagged KDAR candidate (aka, have something that vaguely looks like a <130 MeV muon).
 - Large smearing, absence of a proton track makes angular reconstruction poor for most KDAR events.



Open data: ns timing distributions

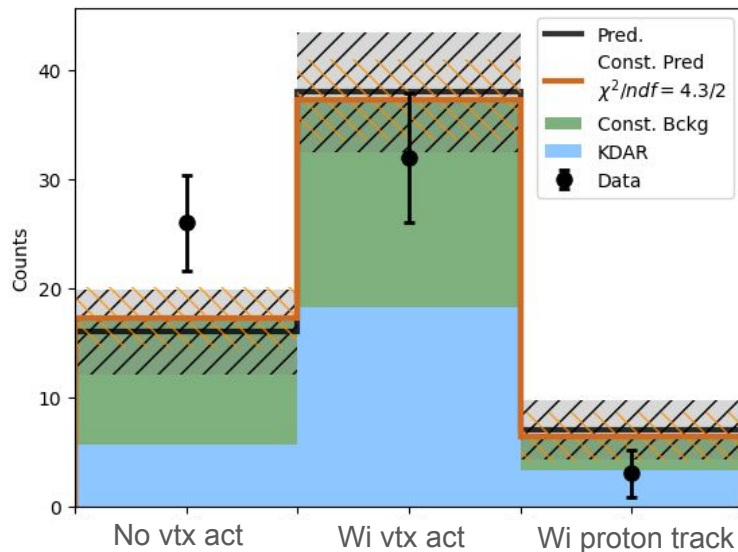
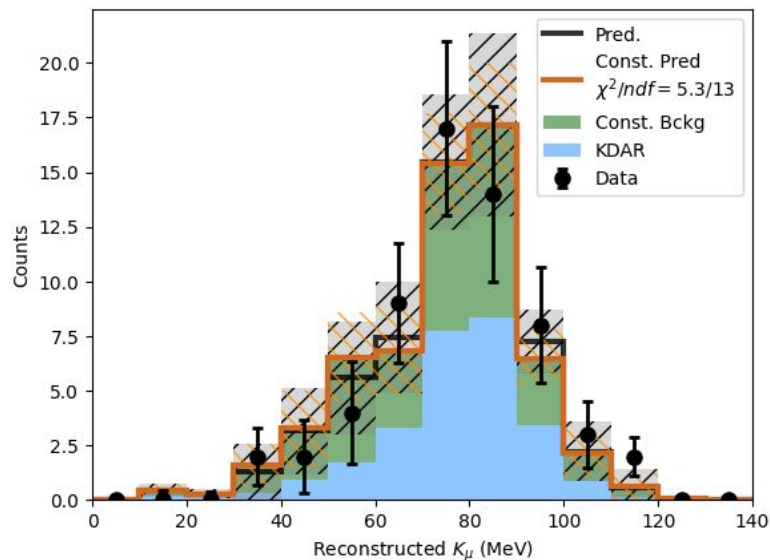
- The observed ns timing distribution is a good validation plot.
 - KDAR should be flatter relative to a more generic selection.
- Events passing the BDT show a flatter ns timing distribution than the preselection.



*subsequent plots cut the background enhanced central four ns timing bins.

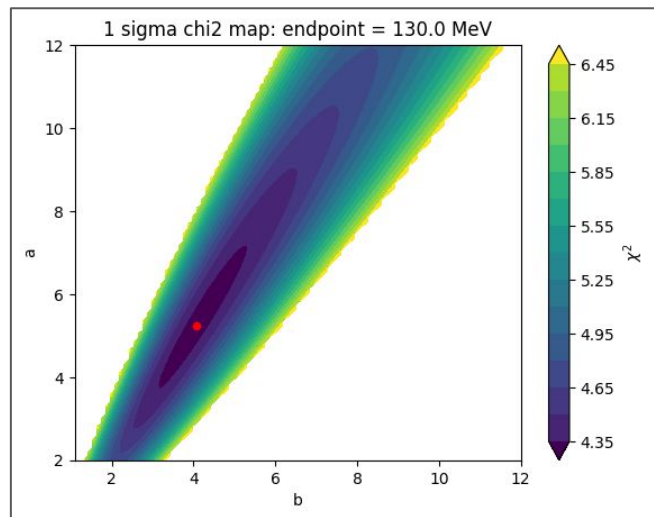
Open data: signal distributions

- No catastrophic data to MC discrepancies observed after BDT and ns timing cuts.
 - The χ^2/ndf is generally less than unity.
- Worst agreement see in plots that include the “proton multiplicity”.
 - First bin is “no vertex activity”, second bin is “with vertex activity”, third bin is “has proton”.
 - [Similar trend](#) in the inclusive selection using an independent set of tools.

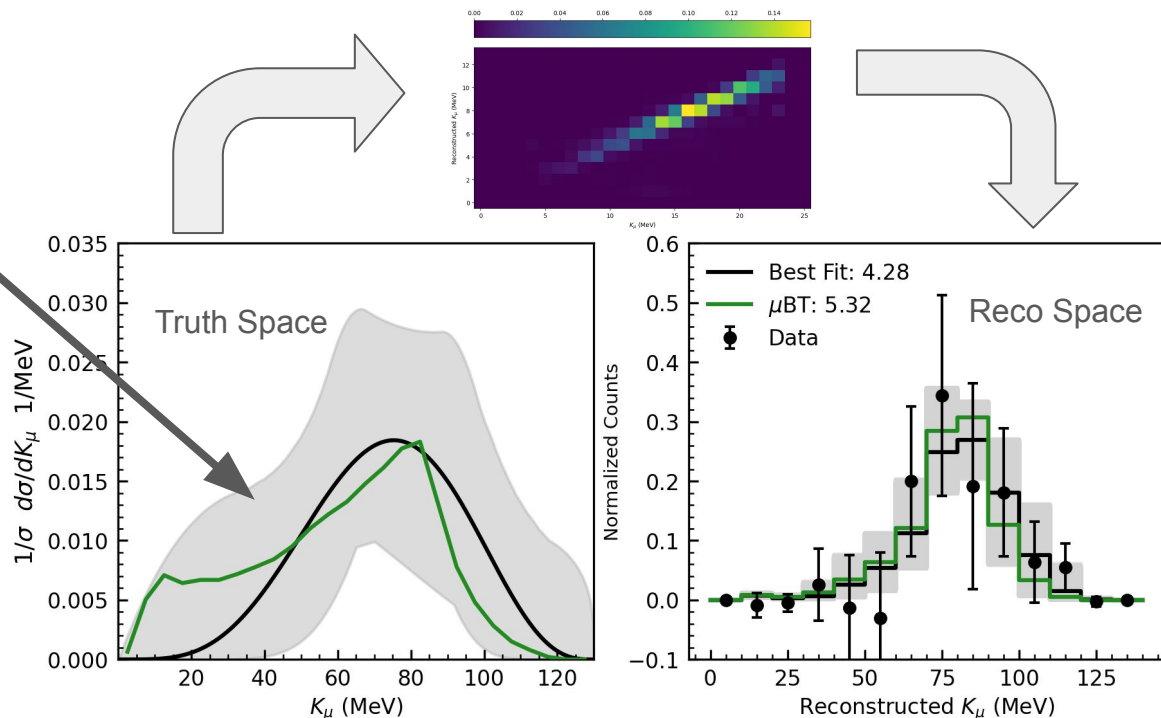


Open data: beta fit

- Large statistical uncertainties prevent the beta distribution fits from converging to a tight region of parameter space.
 - Total uncertainties on the background subtracted data are still large due to the limited statistics.

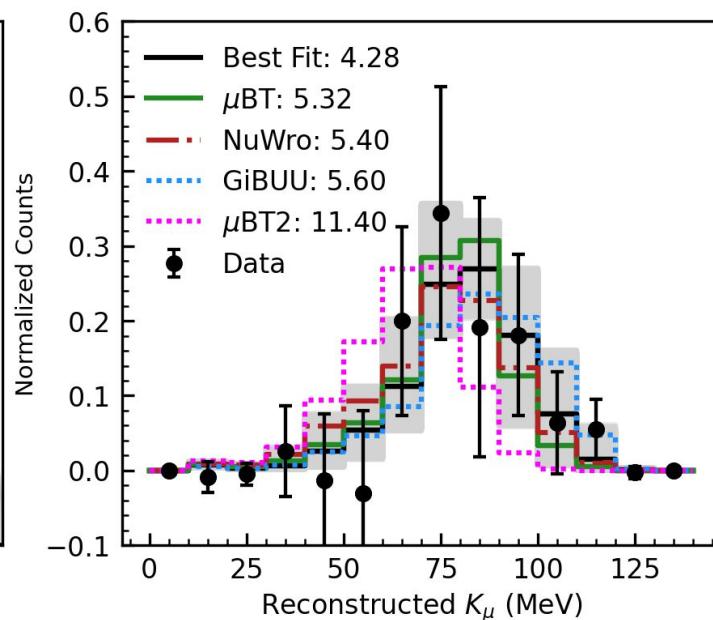
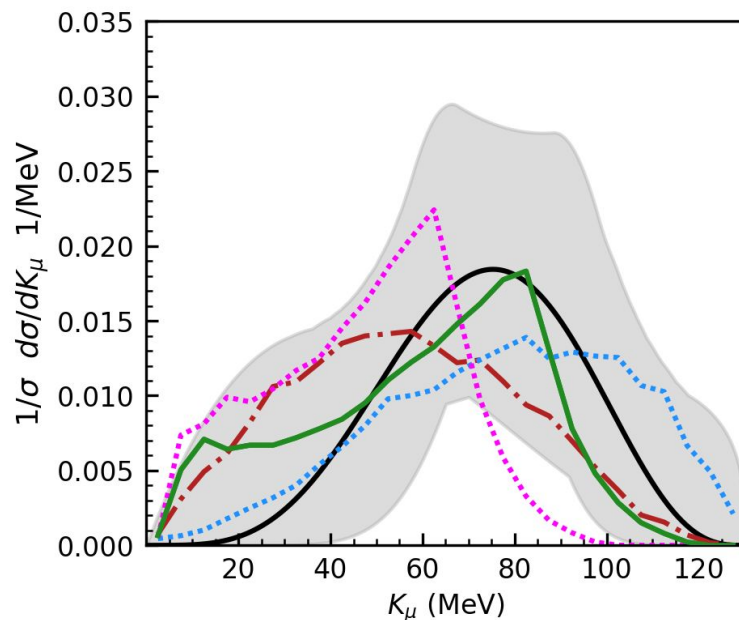


This region defines the bands on the best fit beta distribution (red point) .



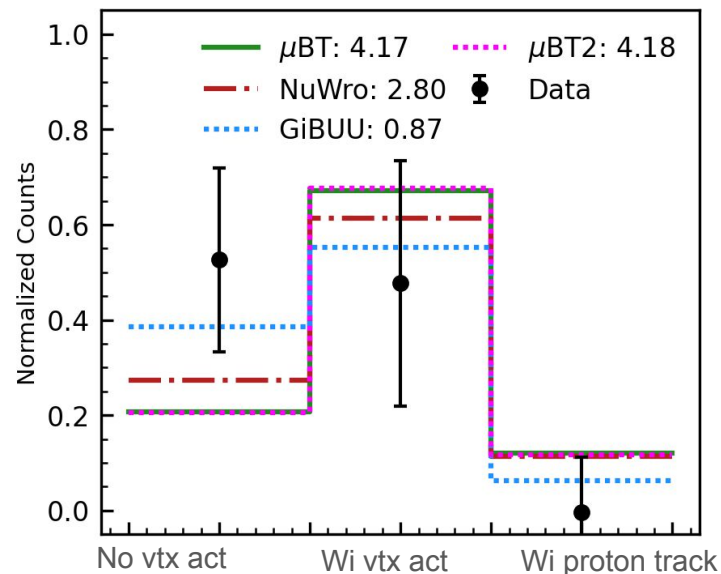
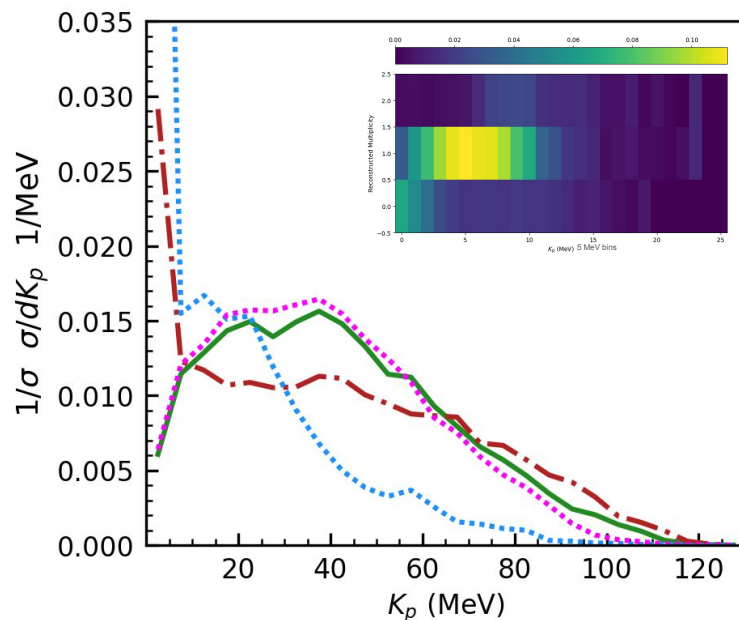
Open data: Kmu generator comparisons

- The Kmu data is shifted to higher energies away from the default MicroBooNE tune GENIE prediction (ubt2).
 - The KDAR prediction for the analysis is utilizing the curve labeled ubt, which is the MicroBooNE tune GENIE prediction with an alternative treatment of binding energy.
- Visible energy ($K_{\mu} + K_p$) shows the same trend.



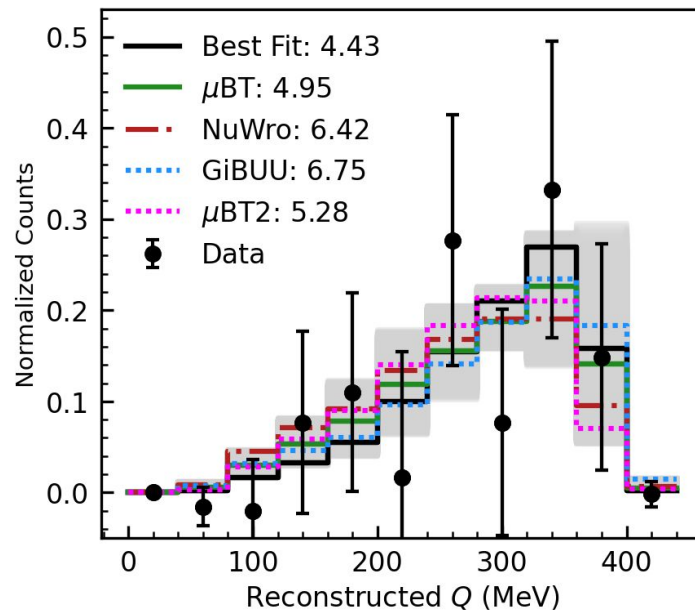
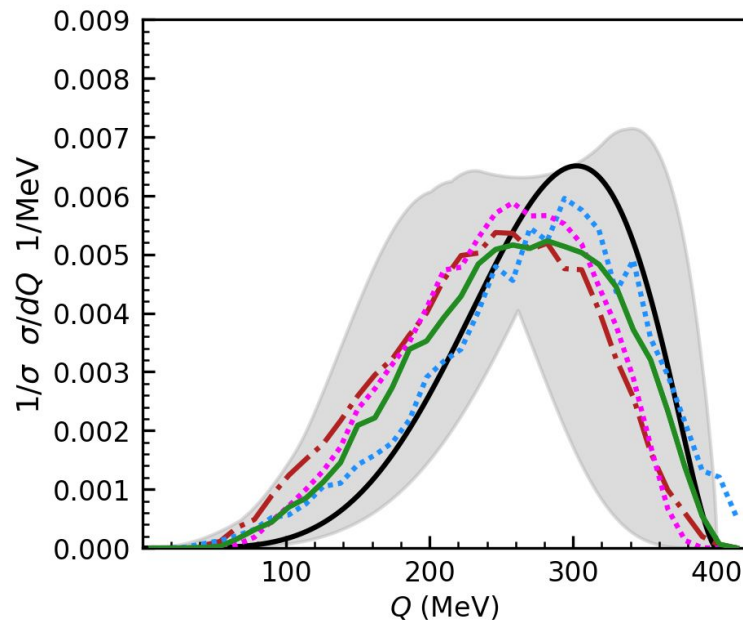
Open data: “multiplicity” generator comparisons

- The proton “multiplicity” distribution shows model preference.
 - No beta fit, proton kinetic energy is folded onto the reconstructed “multiplicity”.
- NuWro and GENIE predict too many events with protons, and not enough without any vertex activity (primarily from sub 15MeV protons).



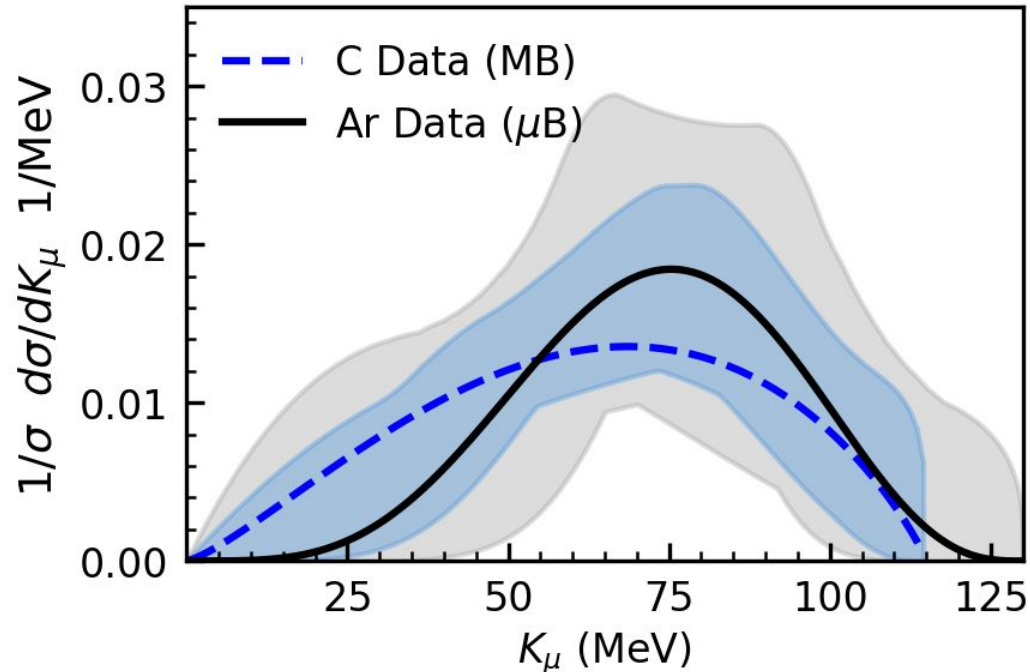
Open data: other generator comparisons

- Still limited discrimination power in most other variables.
 - Muon longitudinal and transverse momentum (p_l and p_t), three momentum transfer (q), four momentum transfer (Q),



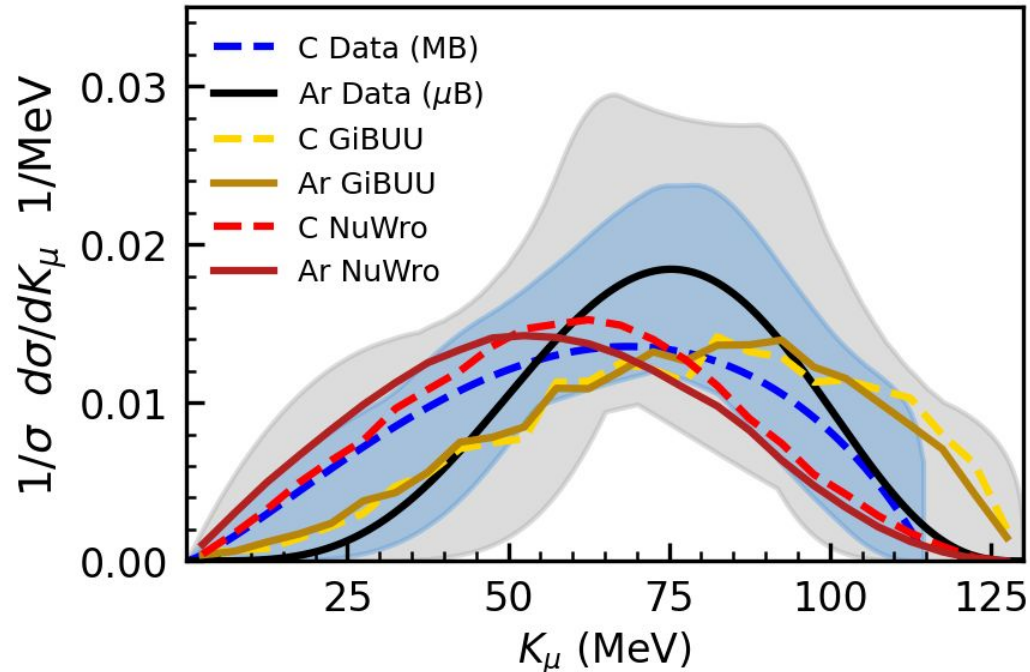
Open data: MiniBooNE Kmu comparison

- [MiniBooNE](#) also measured NuMI absorber KDAR and reported a shape-only beta distribution fit to K_{μ} .
- Unique opportunity to compare C and Ar cross sections at a fixed energy.



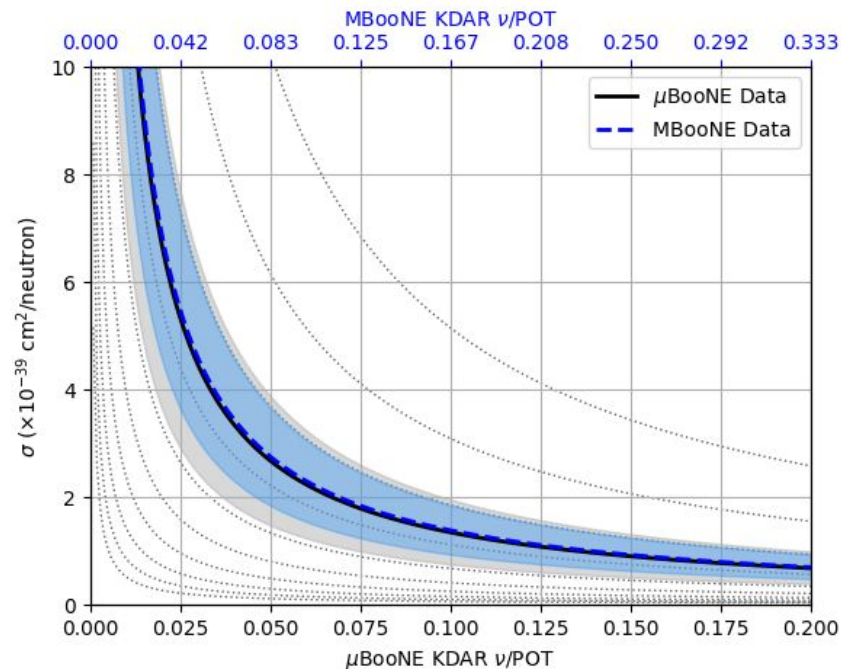
Open data: MiniBooNE Kmu comparison

- [MiniBooNE](#) also measured NuMI absorber KDAR and reported a shape-only beta distribution fit to K_{μ} .
- Unique opportunity to compare C and Ar cross sections at a fixed energy.



Open data: MiniBooNE rate comparison

- We can also compare to the KDAR rate that MiniBooNE measured.
- Presented in a 2D space of cross section and flux.
 - MiniBooNE axis is rescaled by 60% to account for the change from the LE target to the ME target, which is expected to reduce the KDAR rate by this factor.
- Good consistency.



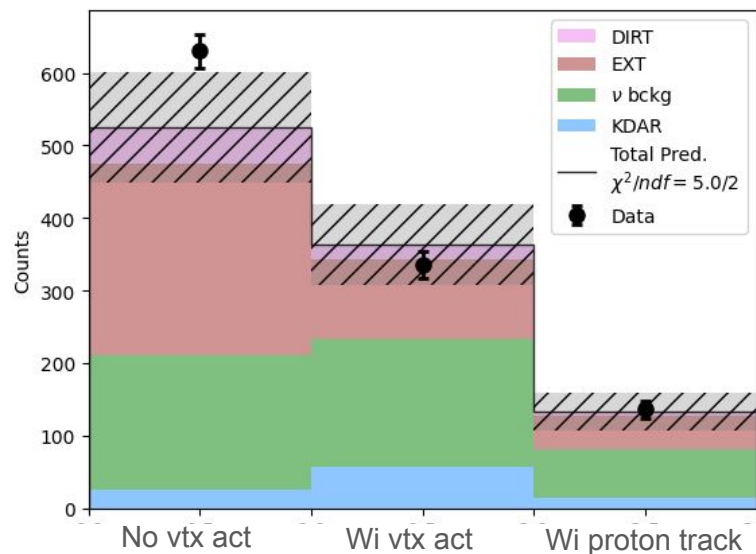
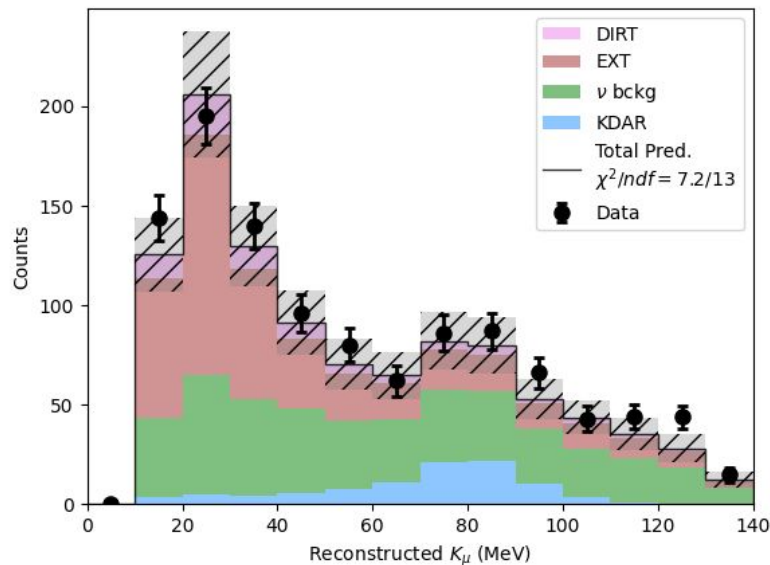
Summary

Many more details in the [internal note](#)

- 9.5e19 open data set unblinded.
 - Split approximately evenly between run 4b RHC data and run 5 FHC data.
- Data favors a K_{μ} (K_p) prediction shifted to higher (lower) energies.
 - K_{μ} disfavors nominal GENIE, K_p favors GiBUU.
 - Limited statistics prevents discrimination power in other variables at this time
- Comparing to the MiniBooNE measurement is an avenue for interpreting the results.
 - Very good consistency observed for the KDAR rate.

Open data: preselection distributions

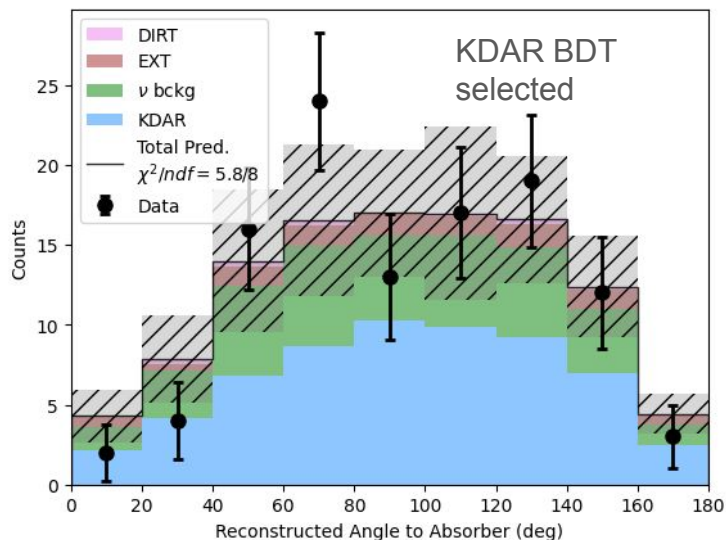
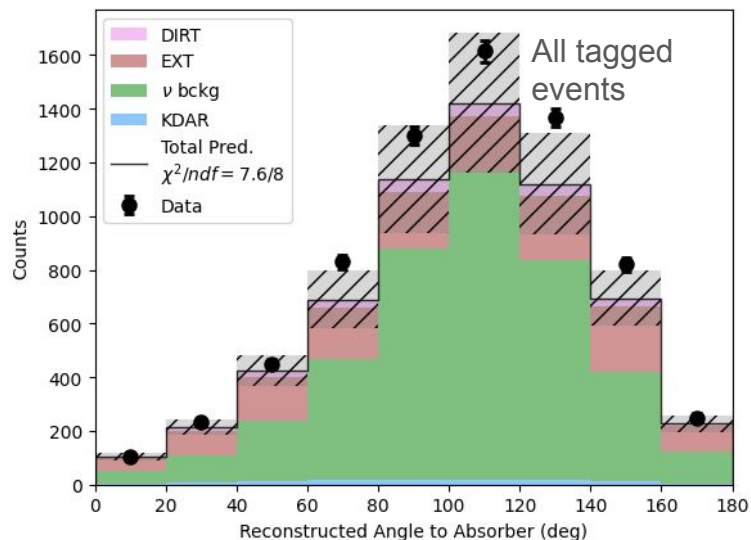
- No catastrophic data to MC discrepancies observed in the preselection.
 - The χ^2/ndf is generally less than unity.
- Worst agreement see in plots that include the “proton multiplicity”.
 - First bin is “no vertex activity”, second bin is “with vertex activity”, third bin is “has proton”.
 - [Similar trend](#) in the inclusive selection using an independent set of tools.



*KDAR prediction is normalized to a “MiniBooNE-like” rate.

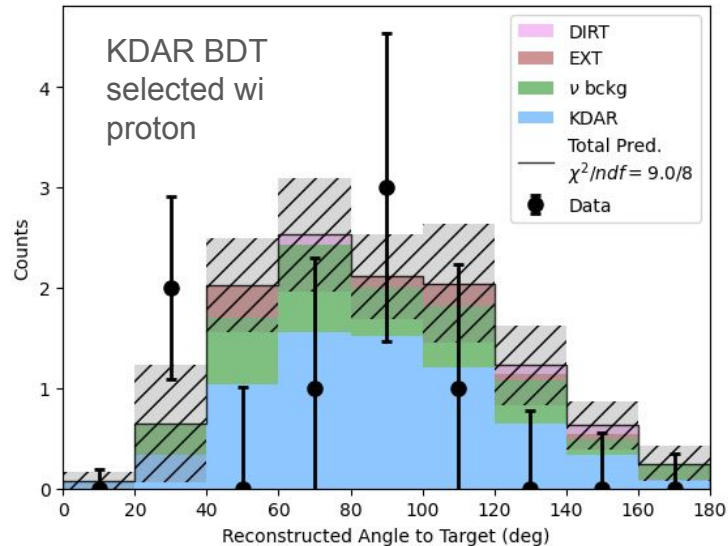
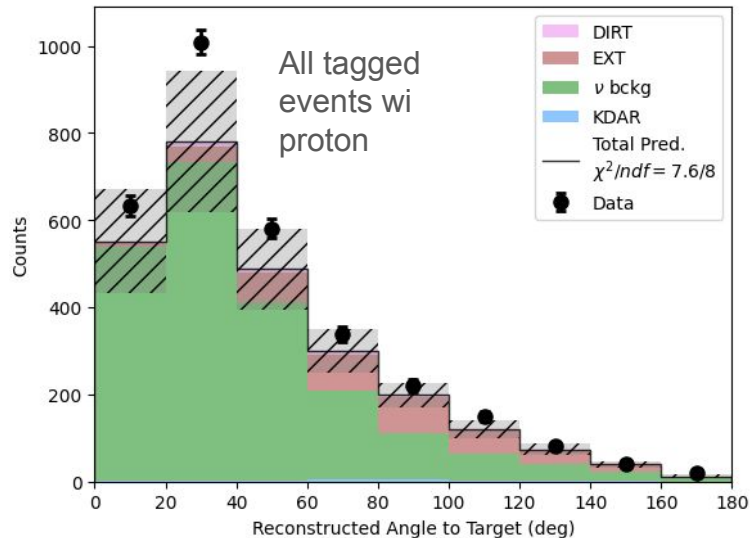
Open data: neutrino angular distributions

- The observed neutrino angular distribution is a good validation plot.
 - KDAR should be shifted away from the target relative to a more generic selection.
- A slight shift towards the absorber is seen when comparing all events with a tagged KDAR candidate (aka, have something that vaguely looks like a <130 MeV muon).
 - Large smearing, absence of a proton track makes angular reconstruction poor for most KDAR events.



Open data: neutrino angular distributions with protons

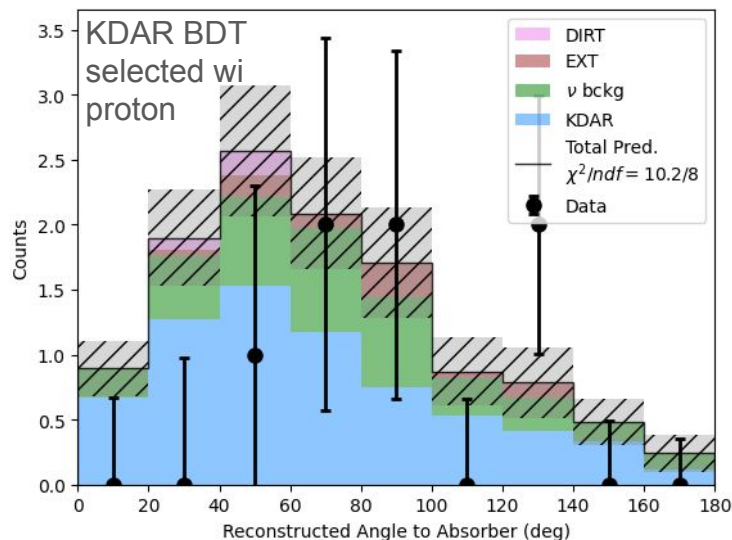
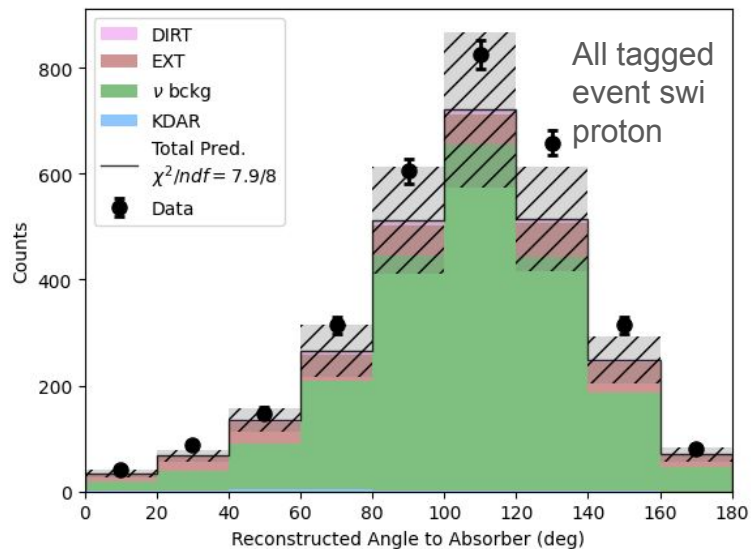
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 - Requiring a proton improves KDAR angular reconstruction, but statistics are much worse.



*no det sys on KDAR BDT due to low stats

Open data: neutrino angular distributions

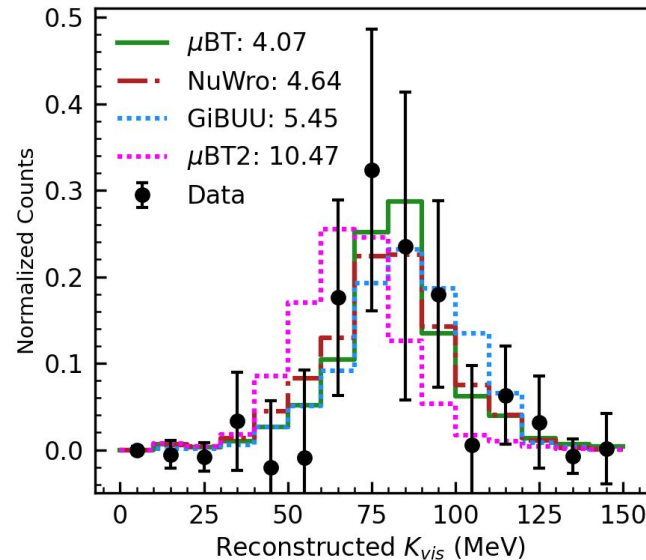
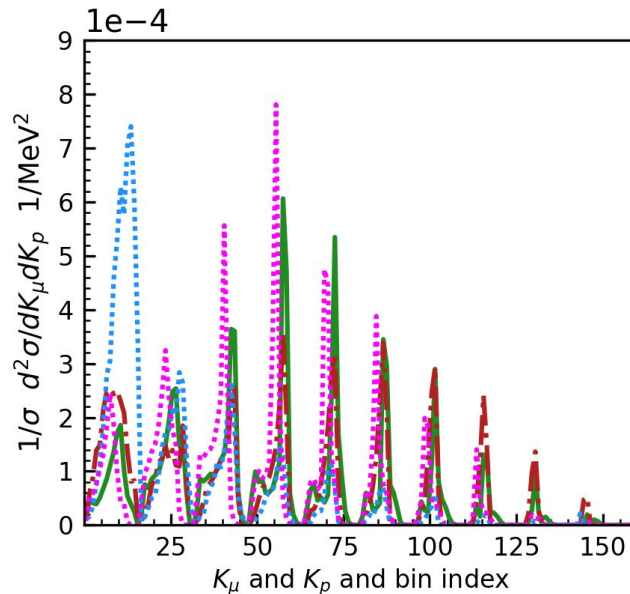
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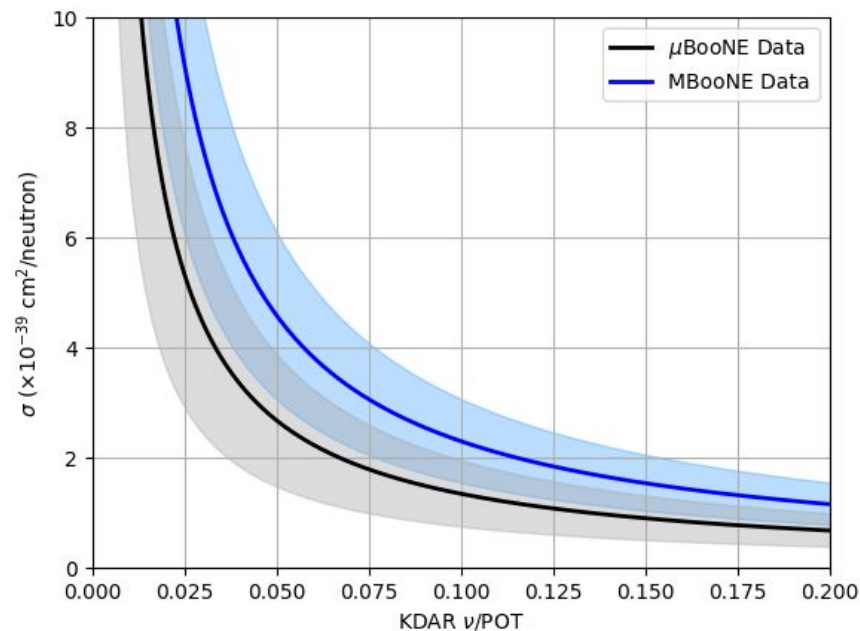
Open data: Kvis generator comparisons

- The Kmu data is shifted to higher energies away from the default MicroBooNE tune GENIE prediction (ubt2).
 - The KDAR prediction for the analysis is utilizing the curve labeled ubt, which is the MicroBooNE tune GENIE prediction with an alternative treatment of binding energy.
- Visible energy ($K_{\mu} + K_p$) shows the same trend.



Open data: MiniBooNE rate comparison

- We can also compare to the KDAR rate that MiniBooNE measured.
- Presented in a 2D space of cross section and flux.
- Need a higher flux at the same XS for MiniBooNE, consistent with the change from the LE target to the ME target.



Form covariance matrix on numuCC and KDAR selections with separate components for sig&bckg in the KDAR selection. The latter has each universe normalized to the CV sig when forming the matrix.

numuCC selection constraints backgrounds in the KDAR selection to get the CV bckg.

Normalize MC signal distributions to the constrained bckg subtracted data. Signal components of cov matrix are normalized by the same factor

- Fold beta distributions through the response matrix and normalize them to the background subtracted event rate.
- Calculate the chi2 between each folded beta distribution and the constrained background subtracted event rate.
- Minimizing the chi2 gives us our best fit beta distribution.
 - Use the delta chi2 to determine the 1 sigma error band on this best fit distribution. This will give us a “truth space” plot reminiscent of the one that MiniBooNE reported.

Fit a beta distribution by minimizing a chi2 calculated from folding into reco space, normalizing, then comparing to data.

Truth space parameterized result used for visualization+interpretation.

Statistical inference done by folding predictions into reco space, normalizing and calculating a chi2.

Xs extraction uses

$$\mathbf{m}_S^{CV} = \phi_S^{CV} + \mathbf{B}_S^{\text{constr}, CV}$$

$$\mathbf{V}_{\mathbf{m}_S \mathbf{m}_S} = \text{Cov}(\mathbf{m}_S, \mathbf{m}_S)$$

$$= V_{\phi_S \phi_S} + V_{\phi_S \mathbf{B}_S}^{\text{constr}} + V_{\mathbf{B}_S \phi_S}^{\text{constr}} + V_{\mathbf{B}_S \mathbf{B}_S}^{\text{constr}}$$

where

$$V_{\mathbf{B}_S \mathbf{B}_S}^{\text{constr}} = V_{\mathbf{B}_S \mathbf{B}_S} - V_{\mathbf{B}_S \mathbf{n}_C} \cdot V_{\tilde{\mathbf{n}}_C \tilde{\mathbf{n}}_C}^{-1} \cdot V_{\mathbf{B}_S \mathbf{n}_C}^T$$

$$\mathbf{B}_S^{\text{constr}, CV} = \mathbf{B}_S^{CV} + V_{\mathbf{B}_S \mathbf{n}_C} \cdot V_{\tilde{\mathbf{n}}_C \tilde{\mathbf{n}}_C}^{-1} \cdot (\mathbf{D}_C - \mathbf{n}_C^{CV})$$

C subscript indicates numuCC control sample and S is KDAR sample. D is data, n/m total rate, ϕ KDAR signal and B background.

“MiniBooNE-like” Approach

Modified GENIE prediction and potential plan moving forward

- The *UseGroundStateRemnant* produces a prediction that covers the relevant phase space without having any obvious “non-physical” aspects.
- **Potential plan moving forward:** Produce the KDAR overlay sample using *UseGroundStateRemnant*, while still just using standard sample with *UseNuclearModel* for the background.
- Take *ValenciaStyleQValue* and *UseNuclearModel* as a unisim for KDAR sample.
 - Former covers small bit of missing phase space at high Emu, latter for consistency with the background prediction.
 - Ideally we would re-run standard samples, but this is too big a production lift, especially because the impact on the background should be much smaller.

